

GLMs in Practice; the Bayesian Thread

Chapters 34 to 38 built the generalized linear model and its inference. This chapter asks two practical questions the theory leaves open. The first is how one arrives at a particular model at all: which family, which link, which scale for the covariates, which interactions, and how candidates are compared. The second is how the model is criticized once fitted, since it has three separable pieces — a mean, a link and a variance function — and a plot that condemns the whole fit is much less useful than one naming the piece at fault.

The rest of the chapter takes up the Bayesian thread, running through the book since [Section 7.6](#). For the normal linear model the posterior was available in closed form and the flat prior reproduced least squares exactly. None of that survives the move to a generalized linear model, for one identifiable reason: the log-likelihood is no longer a quadratic. What replaces it is approximation — the Laplace expansion, which also explains where the BIC comes from — and simulation, in the form of Metropolis–Hastings and Gibbs samplers written out in numpy. The chapter ends where the Bayesian and penalized-likelihood threads meet: a normal prior is ridge regression, a Laplace prior the lasso at its mode, and a merely weakly informative prior is enough to make an estimate exist where maximum likelihood has none.

What you need. [Chapter 34](#) throughout (Definition 34.2.1, Definition 34.2.2, Definition 34.1.2, Theorem 34.3.1, Theorem 34.4.1, Definition 34.5.1, Theorem 34.5.5, Theorem 34.5.1, Definition 34.5.2); [Chapter 35](#) for separation (Theorem 35.3.1), Firth’s penalty (Proposition 35.3.2) and the deviance for binary data (Proposition 35.4.1); [Chapter 37](#) for offsets (Proposition 37.2.1); [Chapter 38](#) for dispersion (Proposition 38.3.2) and overdispersion (Proposition 38.4.1). From earlier parts: [Chapter 20](#) (Definition 20.4.1, Theorem 20.3.1, Proposition 20.1.3); [Chapter 22](#) for Proposition 22.5.1 and Theorem 22.3.2; the parametric bootstrap of Proposition 23.3.3; [Chapter 29](#) (Definition 29.2.2, Proposition 29.2.4, Definition 29.3.1, Proposition 29.5.1); and penalties as priors ([Chapter 27](#), [Chapter 30](#): Definition 27.2.1, Definition 27.4.1, Proposition 30.1.4). The Bayesian thread so far: [Section 7.6](#) for Theorem 7.6.2 and Corollary 7.6.4, [Section 12.6](#) for Definition 12.6.1, and [Section 32.6](#) for Proposition 32.6.1. The diagnostics of [Section 39.2](#) are projection arguments, so they use [Chapter 6](#) directly — the trace of a projection as its rank, the Frisch–Waugh–Lovell theorem and projection in a general inner product (Proposition 6.3.4, Theorem 6.6.2, Theorem 6.9.1) — together with the warning about regressors built from fitted values (Theorem 14.7.2, [Chapter 14](#)).

39.-1.1 Roadmap

- [Building a model](#)
- [Diagnostics](#)
- [Bayesian thread IV: what carries over](#)
- [Bayesian generalized linear models and MCMC](#)
- [Priors as regularizers](#)
- [Summary and notes](#)